# Review: Exponents and radicals

**1** Explain how to evaluate $4^5$.

**2** Explain how to determine if the answer is positive or negative when evaluating an expression with an integer power and negative base such as $(-7)^3$.

**3** By considering the expressions $(-3)^2$ and $-3^2$, state whether the each of the following statements is correct.

**a** Without parentheses, the base is $-3$ and the power is $2$. The result is $9$.

**b** If you have parentheses surrounding the $-3$, then the base is $3$ and the power is $2$. You first evaluate to obtain $9$, then take the negative, which is $-9$.

**c** Without parentheses, the base is $3$ and the power is $2$. You first evaluate $3^2$ to obtain $9$, then take the negative, which is $-9$.

**d** If you have parentheses surrounding the $-3$, then the base is $-3$ and the power is $2$. The result is $9$.

**4** $(-5)^{13}$ simplifies to which of the following?

**A** $5^{-13}$ **B** $-5^{13}$ **C** $-5^{-13}$ **D** $5^{13}$

**5** Evaluate:

**a** $3^1$

**b** $1^3$

**c** $2^6$

**d** $(-2)^4$

**e** $3^3 \times 3^2$

**f** $(-4)^3$

**g** $(-2)^3 \times 3^4$

**h** $-6^3$

**i** $7^2 - 2^2$

**j** $(-11)^2 - (-9)^2$

**k** $(-3)^2 \times (-2)^2$

**l** $-4^3$

**m** $(-3)^4$

**n** $-3^4$

**o** $(-4)^2$

**p** $(-16)^2$

**q** $-15^2$

**r** $-(-8)^2$

**s** $(-9) \times (-6)^2$

**t** $(-2)^3 + 5^2$

**u** $4 - (5 - 2)^2$

**v** $6 - (12 - 3^2)$

**w** $9 - 2^2 \times (-7)$

**x** $(2 - 5)^2 \times (-21 + 14)$

**y** $-(-3)^3 + 8^2$

**z** $-(-4)^3 - (-10)^2 - 122$

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**6** A square patch of grass has a side length of $39\ \text{m}$. Write an expression for the area of the grass in exponential form.

**7** Determine whether each of the following statements is true or false.

**a** $-x^n = (-x)^n$ for all $x$, only if $n$ is odd.

**b** $-x^n = (-x)^n$ for all $x$ always.

**c** $-x^n = (-x)^n$ for all $x$, only if $n$ is even.

**8** Evaluate the following:

**a** $\sqrt{4}$ **b** $\sqrt{16}$ **c** $\sqrt{64}$ **d** $\sqrt{100}$

**e** $\sqrt{144}$ **f** $\sqrt{25} - \sqrt{9}$ **g** $\sqrt{8^2 + 6^2}$ **h** $\sqrt{10^2 - 6^2}$

**i** $\sqrt{144} - \sqrt{16}$ **j** $\sqrt{16}$ **k** $\sqrt{100}$ **l** $\sqrt{169}$

**m** $\sqrt{225}$ **n** $-\sqrt{4}$ **o** $-\sqrt{81}$ **p** $-\sqrt{121}$

**q** $-\sqrt{196}$

## Additional questions

**9** For each of the following, determine the square root.

**a** We know that $5 \times 5 = 25$. Find$\sqrt{25}$.

**b** We know that $7 \times 7 = 49$. Find$\sqrt{49}$.

**c** We know that $6 \times 6 = 36$. Find$-\sqrt{36}$.

**d** We know that $7 \times 7 = 49$. Find$-\sqrt{49}$.

**10** For each of the following, determine the cube root.

**a** We know that $2 \times 2 \times 2 = 8$. What does $\sqrt[3]{8}$ equal?

**b** We know that $4 \times 4 \times 4 = 64$. What does $\sqrt[3]{64}$ equal?

**c** We know that $-3 \times -3 \times -3 = -27$. What does $\sqrt[3]{-27}$ equal?

**d** We know that $-5 \times -5 \times -5 = -125$. What does $\sqrt[3]{-125}$ equal?

**11** Evaluate the following:

**a** $\sqrt[3]{8}$ **b** $\sqrt[3]{64}$ **c** $\sqrt[3]{125}$ **d** $\sqrt[3]{27}$

**e** $\sqrt[3]{-1}$ **f** $\sqrt[3]{-27}$ **g** $\sqrt[3]{-8}$ **h** $\sqrt[3]{-64}$

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**12** Which is bigger: $3^3$ or $5^2$?

**13** Evaluate the following expressions:

**a** $\left(2^2+3^3\right)$ **b** $(3+4)^2$ **c** $2^3(3+7)$ **d** $\left(\sqrt{9}+1\right)^3$

**e** $\left(\sqrt{3^2+4^2}\right)^3$ **f** $\dfrac{2^3+\sqrt{16}}{4}$ **g** $\sqrt{64}\div\sqrt[3]{8}$ **h** $2\left(\sqrt[3]{8}+4\right)^2$

**i** $\left(\dfrac{1}{2}\right)^2+\dfrac{1}{2}$ **j** $2\left(\dfrac{2}{3}\right)^2$ **k** $\left(\dfrac{2}{4}+\dfrac{1}{2}\right)^2$ **l** $\left(\dfrac{2}{3}+\sqrt{\dfrac{1}{9}}\right)^3$

**m** $\sqrt{1-\left(\dfrac{6}{10}\right)^2}$ **n** $\left(-\dfrac{2}{3}\right)^2+\sqrt{\dfrac{25}{81}}$ **o** $\sqrt{81}\div\sqrt[3]{125}$ **p** $2\left(\dfrac{1}{\sqrt[3]{8}}+4\right)$

**14** For each of the following situations:

**i** Determine the error each student made.

**ii** Write the correct work and answer.

**a** Ralph tried to evaluate the expression $2(3+2)^2$. He thought that he had the right answer and submitted the following to his teacher $2(3+2)^2=2(5)^2=10^2=100$.

**b** Ida tried to evaluate the expression $\sqrt{9}\left(8\div4-3\times4+4^2+3\right)$. She submitted the following to her teacher $\sqrt{9}\left(8\div4-3\times4+4^2+3\right)=\sqrt{9}(2-12+16+3)=\sqrt{9}(9)=\sqrt{81}=9$.

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